Conceptual similarity, but not pragmatics, shapes evidential systems during learning

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Languages are shaped by pressures for simplicity and informativeness (e.g., Carr et al., 2020; Kemp et al., 2018); but how pragmatics plays into this dynamic has not been fully investigated. Here, we investigate this question in the domain of evidential systems. Evidential marking falls on a continuum of directness, ranging from direct (perceptual) evidence type to indirect (reportative) evidence type, with inferential evidence type in between. All languages with grammatical evidential marking explicitly mark reportative evidence, and the majority mark *only* that type of evidence (Aikhenvald, 2018). Further, no language has marking for direct and reportative without also marking inferential evidence. Languages are thus sensitive to the continuum of directness, with more indirect sources of evidence more likely to be marked. This may reflect a pragmatic pressure (Saratsli & Papafragou, 2023) related to accountability: flagging a distinct information source may deflect speaker responsibility (Mahr & Csibra, 2021). However, evidential systems may also be shaped by a simplicity pressure for marking conceptually similar categories together. Recent work on the learning of evidential systems (Saratsli et al., 2020) does not de-confound these two potential pressures; thus, our experiment aims to do so.

We train English-speaking participants on an evidential system expressing two evidence types (either marked distinctly, or one unmarked), with the third type held out (Fig. 1). We measure accuracy and extrapolation choices for the held-out type. Across four conditions (N=60 each), we find no learning preference for marking indirect evidence (Fig. 2A); however, a binomial logistic regression reveals lower accuracy on a system that disrupts the conceptual grouping of direct and inferential evidence (infer. X, direct ∅: β =-0.34±0.08, $p < .001$).

Figure 1. Example training (left, middle) and test (right) trials. Participants learn a direct ('bo') and a reportative ('di') evidential marker, then must extrapolate one of them to inferential evidence.

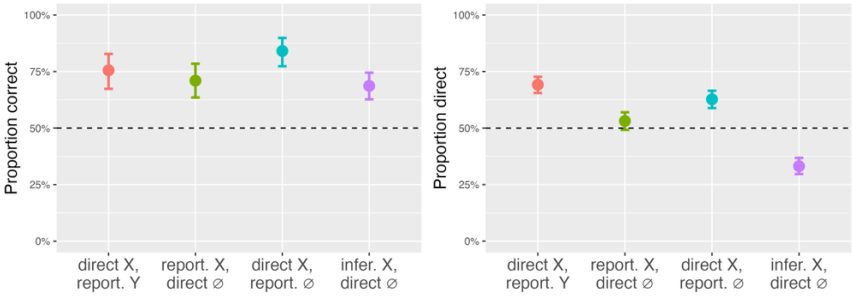

Figure 2. A: Average accuracy on trained evidence types by condition. B: Average proportion extrapolation of direct marker to held-out evidence type by condition. Error bars are 95% confidence intervals. X and Y indicate distinct markers, ∅ indicates unmarked.

In three conditions we also find a preference to group direct and inferential evidence types separately from reportative evidence type (Fig. 2B). Participants extrapolate a learned marker from direct to inferential (direct X, report. Y: $p < .001$, 69.2%, 95% CI [65.5%, 72.7%]; direct X, report. ∅: $p < .001$, 62.7%, 95% CI [58.8%, 66.5%]), but not from reportative to inferential (report X, direct ∅: $p = .119$, 46.9%, 95% CI [43.0%, 50.8%]) evidence type. In the final condition (infer. X, direct ∅) participants extrapolate the inferential to the reportative evidence type ($p < .001$, 66.8%, 95% CI [63.2%, 70.3%]), preserving the continuum of evidential directness.

Overall, our results suggest that during learning, a pressure for marking conceptually similar meanings together shapes evidential systems. By contrast, we find no evidence for an influence from pragmatics on marking. These findings align with previous research showing that language is shaped by distinct pressures during learning and communication: learning favors simplicity or category naturalness; communication may favor other pressures like informativeness (Kirby et al., 2015; Smith & Culbertson, 2025) and accountability.

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